LAURA RINALDI

Piove di Sacco (PD), 27/07/1996

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SCIENTIFIC INTERESTS

- ***Continuum Mechanics*** *— modelling of complex materials and multiphysics interactions*
- ***Surrogate Modelling & Operator Learning*** *— reduced-order models, data-driven PDE solvers*
- ***Digital Twins*** *— real-time computational frameworks for monitoring and prediction*
- ***Uncertainty Quantification*** *— forward/ inverse UQ, sensitivity analysis*

Applications

- ***Biological Systems*** *— bread leavening and thermal processing, soft-matter behavior*
- ***Fluid Dynamics*** *— laminar, contraction flows, flow past a cylinder*
- ***Cancer-on-Chip Models*** *— coupled Keller–Segel-type systems*

RESEARCH POSITIONS

2025 - today — **Junior Researcher** at the Institute of Applied Mathematics and Information Technologies "E. Magenes" (IMATI), Pavia, contributing to the project “Digital Driven Diagnostics, prognostics and therapeutics for sustainable Health care” D34 Health – PNC 0000001 - CUP B53C22006100001 – Spoke 3.

2025, 11 mos. — **Postdoctoral researcher** at the University of Padua, contributing to the project “Tchakaloff-Davis-Wilhelmsen measure compression and applications to numerical modelling”.

EDUCATION

31-03-2025 — **PhD in Mathematical Sciences** at the University of Padua, with a doctoral thesis on mathematical physics, modeling, numerical analysis, and machine learning. The thesis, titled “*A mathematical construction of the digital twin of bread leavening*”, was developed within the framework of the project “Reduction of energy waste in food and beverage preparation activities”. SOCINT–G-Research Award for the **best PhD thesis** in quantitative disciplines**.**

25-09-2020 — **Master’s Degree in Mathematics** at the University of Padua with 110/110 with honors and thesis in the physical-mathematical field “*Bose Hubbard model of cold atoms to higher orders*”.

28-09-2018 — **Bachelor’s degree in Mathematics** at the University of Padua with 100/110 and thesis in the physical-mathematical field entitled “*Shading properties and topological stability for discrete dynamical systems*”.

2015 — High School Diploma at “Istituto d'istruzione superiore A. Einstein”, Scientific High School - Piove di Sacco.

PARTICIPATION TO SCHOOLS AND COURSES

May 2025	Participating in Virtual-exchange course: "Learning-enhanced Data Assimilation" (4 CFU), promoted by the University of Padova and the University of Southern California (USA).
Jul 2024	Participating in Udine CISM "Advanced Course on Inverse Problems for Mechanical Systems: Methods, Simulations and Experiments".
Jul 2023	Participating in the "Barcelona XX Jacques-Louis Lions Spanish-French School on Numerical Simulations in Physics Engineering". **Best poster** presented "Bread leavening and baking: a modelling challenge".

PROFESSIONAL WORK EXPERIENCES

2021, 10 mos.	Apprenticeship at the "SCAI ITEC S.r.l." company in Padua with database administration duties.
2020, 4 mos.	Internship at the company "SCAI ITEC S.r.l." in Padua with training objectives in database administration.
2020, 1 mo.	"Ready to Oracle" training course to learn Oracle Database, SQL Server and SQL Language.
30-03-2021	Oracle Database SQL Certified Associate 1Z0-071.

TEACHING ACTIVITIES

2025 - 2026	**Teaching** of "Istituzioni di matematica con esercitazioni – mod.1" for the degree course in Environmental Science at the University of Ca' Foscari (Venice). Teacher (48 h).
2025, 4 mos.	**Supplementary teaching activity** of "Calcolo Numerico" for the degree course in Energy Engineering, Prof. Alvise Sommariva at the University of Padua. Teacher support activities during laboratory with MATLAB exercises (24 h).
2022, 4 mos.	**Supplementary teaching activity** of "Calcolo Numerico" for the degree course in Mathematics, Prof. Michela Redivo-Zaglia at the University of Padua. Teacher support during laboratory with Python/MATLAB scripts and exercises (25 h).
2021, 4 mos.	**Supplementary teaching activity** of "Fondamenti di Analisi Matematica 2" for the degree course of Chemical and Materials Engineering (nr. Channel 1), Prof. Luca Baracco at the University of Padua. Exercises and solutions of past exams (24 h).

MENTORING EXPERIENCE

19-09-2025	Co-supervisor of M. Storgato. **Bachelor's thesis** "*Quadratura subperiodica in Python"*.
19-09-2025	Co-supervisor of G. Traversin. **Bachelor's thesis** "*Implementazione in Python di un algoritmo per la cubatura numerica su elementi poligonali con un lato curvo"*.
11-04-2025	Co-supervisor of E. Caregnato. **Master's thesis** "*Goal Oriented Operator Networks"*.
19-04-2024	Co-supervisor of M. Dell'Orto. **Master's thesis** "*Variationally mimetic operator neural networks for the time-dependent heat equation"*.

AWARDS AND PRIZES

2024	Winner of the "*SOCINT- G-Research Award"* for **the best PhD thesis** in quantitative subjects – II Edition.
2024	**Third place** in the "Three-Minutes Thesis Competition (3MT)" at the University of Padua, promoted by Coimbra Group.
2023	Winner of the **best poster** presented at the "XX Jacques-Louis Lions Spanish-French School on Numerical Simulations in Physics Engineering".
2023	Winner of the **best presentation** among PhD students at the conference "FEniCS 2023".

GRANTS AND SCHOLARSHIPS

2026	The ESMINT Pilot Research Grant "ASPIRA-3D. Influence of collateral circulation and catheter caliber on aspiration thrombectomy efficacy: in vitro and in silico study" (10000,00 €).
2026	GNCS-INDAM Project "Metodi computazionali avanzati per sistemi complessi: approcci ibridi numerici e machine learning" (1700,00 €).
2023	GNCS-INDAM grant for participating in the "SIMAI 2023" conference in Matera.
2018	Regional scholarship winner.
2018	Grant for participating in the "Junior Math Days" workshop at the SISSA University of Trieste.
2017	Regional scholarship winner.
2016	Regional scholarship winner.

AFFILIATIONS TO MATHEMATICAL GROUPS AND PROFESSIONAL SOCIETIES

Since 2023	Member of the "Società Italiana di Matematica Applicata e Industriale" (SIMAI).
Since 2023	Early Career Membership of the "Society of Industrial and Applied Mathematics" (SIAM). Before 2026 (student membership).
Since 2022	Member of the "Gruppo Nazionale Calcolo Scientifico" (GNCS) of the Istituto Nazionale di Alta Matematica (INdAM).
2022	Member of "Unione Matematica Italiana" (UMI).

CONFERENCES AND SEMINARS

Oct 2026	Pavia IMATI – IGM MEETING Talk "Uncertainty quantification for organs-on-chip" – **invited**.
Jul 2026	Munich WCCM ECCOMAS 2026. Talk "*UQ analysis for cancer-on-chip experiments*".
Jul 2026	Graz ECMTB 2026. Talk "*UQ analysis for cancer-on-chip experiments*" – **invited**.
Mar 2026	Pavia IMATI-CNR. Seminari di Matematica Applicata. Talk "*The role of surrogate models in different applied cases: bread leavening and fluid dynamics*" – **invited**.
Sep 2025	Trieste SIMAI 2025. Talk "*Building a surrogate model for coupled, time-dependent PDEs using Operator Networks for thermal applications*" – **invited**.
Apr 2025	University of Padua. Mathematical Physics and Dynamical Systems Seminar. Talk "*A mathematical model of bread leavening for building a digital twin*" – **invited**.
Aug 2023	Matera SIMAI 2023. Talk "*On approximating fictitious heat sources arising in an inverse heat transfer problem*" – **invited**.
Jun 2023	Pula FEniCS 2023. Talk "*Bread baking simulation with FEniCS*".
Sep 2022	Arenzano YAMC Second Conference of Young Applied Mathematicians. Talk "*Modelling for bread preparation to avoid energy waste*".
Sep 2022	Pavia GIMCS SIMAI Young 2022. Talk "*Modelling for bread preparation to avoid energy waste*".
May 2020	University of Padova. Talk "*The quantum harmonic oscillator: analysis and resolution*".

WORKSHOPS

Jun 2026	University of Milano Bicocca. 8th Lombardy Young Numerical Analysts Meeting (LYNUM-VIII). Talk "*Surrogate-model-based uncertainty quantification for cancer-on-chip experiments*" – **invited**.
May 2025	University of Pavia. 7th Lombardy Young Numerical Analysts Meeting (LYNUM-VII).
Feb 2025	Padova DTW 2025 International Workshop on Digital Twins: Mathematical Analysis, Formal Methods and Scientific Machine Learning. Poster "*Digital twin of bread leavening for energy saving*".
Nov 2024	Milano M4E: Mathematics for Planet Earth. Poster "*Digital twin of bread leavening for energy saving*".
Dec 2018	SISSA University of Trieste. Workshop "Junior Math Days".

SCIENTIFIC PUBLICATIONS

IN PREPARATION

1. **L. Rinaldi,** M. Pennacchio, L. Tamellini. "HDG-based approximation and VarMiON-based prediction for the two-dimensional Keller–Segel system"
2. **L. Rinaldi,** S. Bertoluzza, G. Bretti, L. Tamellini, P. Zanotti. "Advanced Uncertainty Quantification in Complex Cancer-on-Chip Systems with Graph and Maze Geometries "

PUBLISHED

1. **L. Rinaldi,** G. G. Giusteri. "Variationally mimetic operator network approach to transient viscous flow". Advances in Computational Science and Engineering Vol. 10, 2026 doi:10.3934/acse.2026015

2. **L. Rinaldi,** A. Sommariva, M.Vianello. "*ORTHOCUB: integral and differential cubature rules by orthogonal moments*". Journal of Approximation Software, 3(2), (2026) https://doi.org/10.13135/3103-1935/12995

3. **L. Rinaldi,** A. Sommariva, M.Vianello. "*Effective numerical integration on complex shaped elements by discrete signed measures",* Finite Elements in Analysis and Design, Volume 254, (2026) https://doi.org/10.1016/j.finel.2025.104505

4. E. Chinellato, P. Martin, **L. Rinaldi**, F. Marcuzzi. "*Exploiting scientific machine learning on embedded digital twins*", Springer series: Lecture Notes in Computational Science and Engineering - Math to Product (2025), https://link.springer.com/chapter/10.1007/978-3-031-95709-3_12

5. **L. Rinaldi,** G. G. Giusteri. "*Modelling and simulation of bread leavening to monitor energy consumption in industrial processes*", Commun. Appl. Ind. Math. 16(1), 41-61 (2025) https://doi.org/10.2478/caim-2025-0003

6. G. G. Giusteri, F. Marcuzzi, **L. Rinaldi**. "*Replacing voids and localized parameter changes with fictitious forcing terms in boundary-value problems*", Results Appl. Math. 20, 100402 (2023) https://doi.org/10.1016/j.rinam.2023.100402

DOCTORAL THESIS

7. **L. Rinaldi**. "*A mathematical construction of the digital twin of bread leavening*". Ph.D. Thesis (2025) https://tesidottorato.depositolegale.it/handle/20.500.14242/218031

REVIEWER FOR

2026 Young Applied Mathematicians Conference, 6th edition.

LANGUAGES

Italian: mother tongue.
English: good knowledge of spoken and written language, B2 certified at the University Language Center (CLA) of the University of Padua.
French: school level.

COMPUTER SKILS

Coding: Matlab/Octave (particularly, Sparse Grids MATLAB kit, Umbridge), Python (particularly, FeniCS/FEniCSX libraries for solving partial differential equations, PyTorch for deep learning), LATEX (Beamer for presentation), Wolfram Mathematica, SQL query language
Database: Oracle, SQL Server
Other tools: KNIME, SQL Developer, Toad, PuTTY, FileZilla, and WinSCP, Git/GitHub, Docker
Software: ParaView, Microsoft Office
OS: Linux/Ubuntu, Windows

SOFTWARES

A code repository is available on my GitHub page: laura-rinaldi (Laura Rinaldi) · GitHub

ORGANIZATION AND EVENTS

2023-2025	Organizer duties for the “Terza Missione” activity **Science4all:** a volunteering initiative promoted by the University of Padua, designed for primary school students aged 8 to 11. The program offers interactive workshops and hands-on science labs that aim to spark curiosity, foster critical thinking, and inspire a love for science in a fun and engaging way.

EXTRACURRICULAR INTERESTS

Practiced at the amateur level in figure skating, speed skating, and ice skating.
License as a volleyball referee manager obtained from the provincial committee FIPAV of Padua.
Certificate of attendance in the piano course.
Practiced activities with the task of assistant-educator for children aged 3 to 6 years.

In compliance with the Italian legislative Decree no. 196 dated 30/06/2003, I hereby authorize you to use and process my personal details contained in this document

Padova, 14/10/2026

Laura Rinaldi